

A Large-Scale Multi-Species Leaf Disease Dataset for Benchmarking Self-Supervised Learning

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Abstract—Plant disease recognition is important for precise agriculture but we rely on large labeled datasets limits scalability in real world scenarios. This work introduces a large scale leaf image dataset and shows its effectiveness for self-supervised learning. The dataset contains images from four plant species Camphor(*Cinnamomum Camphora*), Haritaki(*Terminalia Chebula*), Sojina (*Moringa Oleifera*), Neem (*Azadirachta Indica*) covering five leaf conditions: Healthy, Bacterial Spot, Shot Hole, Yellow and Powdery Mildew. The dataset comprises thirteen final classes. A total of 10,858 images were using multiple mobile cameras and expanded to 65,148 samples through data augmentation. Since most plant disease research studies use less than 2,000 images our dataset can be considered as a relatively large scale dataset within agricultural datasets. All images were standardized to 512×512 resolution. We evaluated multiple self-supervised learning frameworks in our dataset which are MoCo, BYOL and DINO through linear probing, MLP, k-NN classifier and fine tuning on downstream classification tasks. Our experiment shows that SSL pretrained models consistently outperform the baselines and produce strong label efficiency which highlights the dataset's suitability for self-supervised learning. This study supports scalable plant disease detection and advances automated plant health monitoring systems.

Index Terms—Self-supervised learning, plant disease detection, leaf image classification, agricultural computer vision, MLP, Linear-probing, k-NN classifier, MoCo, BYOL, DINO

I. INTRODUCTION

Plant diseases can be a significant threat to agricultural productivity and food security worldwide. Early and accurate disease detection promise better efficiency in overall crop production. Recent studies in deep learning have shown strong performance in automated leaf disease classification but most approaches depend heavily on a fully labeled datasets. Acquiring expert level annotations is costly, time-consuming and often infeasible at scale.

Traditional supervised deep learning models are commonly used for plant leaf disease classification when sufficient labeled data is available. We did use several ImageNet pretrained architectures which were VGG16, VGG19, DenseNet-201, ResNet-50, InceptionV3, MobileNetV2, NASNet and EfficientNetB5 and they were evaluated to establish a supervised baseline. The results show that model performance improves with increased training data which highlights the strong dependence of supervised learning on labeled samples. InceptionV3 achieved the best performance with a peak accuracy of 98.13% and VGG19 also showed stable and high accuracy of up to 97.51%. Lightweight models such as MobileNetV2 and

EfficientNetB5 performed comparatively worse under limited data conditions. Supervised pretrained models can achieve high accuracy but their reliance on large labeled datasets motivates the adoption of self-supervised learning methods to reduce annotation requirements.

Self-supervised learning (SSL) has emerged as a promising alternative by enabling models to learn rich visual representations from unlabeled data. Methods such as MoCo, BYOL and DINO have shown remarkable success in natural image domains while achieving competitive downstream performance with significantly fewer labeled samples. The application and systematic evaluation of SSL in plant disease analysis remain underexplored on real-world datasets in multiple species and disease types.

That's why we presented a large scale multi-species leaf image dataset collected and designed to support both supervised and self-supervised learning.

II. LITERATURE REVIEW

The application of deep learning to plant disease detection has received increasing attention due to its potential to automate crop monitoring and reducing reliance on manual inspection. Traditional supervised learning approaches have achieved high accuracy in leaf disease classification but their effectiveness largely depends on the availability of large and labeled datasets. Collecting expert labeled data is expensive and time consuming so it is the exploration of alternative learning that can leverage unlabeled data.

Ren *et al.* [1] provided a comprehensive survey of self-supervised learning techniques in recommendation systems. Their work systematically categorized SSL methods into contrastive, generative and predictive learning frameworks and highlighted how auxiliary self-supervised objectives improve representation quality and robustness. Many of the core principles such as representation consistency, invariance learning and label efficiency are directly applicable to computer vision problems. The authors emphasized that SSL is particularly valuable in scenarios where labeled data is limited or expensive a usual case scenario for agricultural image analysis and plant disease diagnosis.

Ferdous *et al.* [2] introduced AI-MedLeafX which is a large scale computer vision dataset designed for medicinal plant diagnosis. The dataset provides high quality, real-world images collected under diverse environmental conditions, finding out

common issues such as background clutter and illumination variability. Their work highlights the importance of dataset scale and diversity for building reliable plant diagnosis systems. The dataset was primarily evaluated using supervised learning approaches which leaves the opportunities to explore how self-supervised pretraining can further enhance performance and reduce annotation dependency.

Güldenring and Nalpantidis [3] investigated the potential of self-supervised contrastive learning on agricultural images. Their work showed that models pretrained using contrastive objectives can learn meaningful representations from unlabeled data and subsequently outperform randomly initialized networks on downstream agricultural tasks. The study emphasized the importance of carefully selected data augmentations that preserve semantic information while encouraging representation invariance. The experiments were primarily limited to contrastive methods and did not focus on multi-species leaf disease classification or large-scale class diversity.

Wang *et al.* [4] explored the role of self-supervised learning in plant leaf disease recognition. They proposed an SSL-based framework and evaluated its performance under varying amounts of labeled data. Their findings showed that SSL-pretrained models achieved higher classification accuracy and stronger generalization compared to fully supervised baselines. This study highlighted the label efficiency of SSL approaches but the dataset used in their experiments was relatively controlled with limited variation in acquisition conditions and plant species and which may restrict real-world applicability.

A broader overview of SSL applications in plant disease detection was presented by Al Mamun *et al.* [5] in a recent review. The authors categorized existing methods into contrastive, clustering and reconstruction based approaches and analyzed their strengths and limitations. The review identified several open challenges like dataset bias, insufficient evaluation across multiple plant species and the lack of standardized benchmarks for fair comparison. The authors emphasized the necessity of diverse, large-scale datasets and comprehensive experimental protocols to fully assess the robustness of SSL methods in agricultural environments.

Anwar and Masood [6] proposed VitNas which is a Vision Transformer based deep learning model for efficient medicinal plant identification using real-world datasets. Their study showed that transformer architectures can effectively capture global visual patterns and outperform traditional convolutional models in plant classification tasks. VitNas achieved strong results under supervised settings but the reliance on labeled data remains a limitation when scaling to larger or more diverse plant species collections.

The present work shows several limitations identified in prior studies. We introduce a large-scale, multi-species leaf disease dataset collected under real-world conditions and evaluate multiple state-of-the-art self-supervised learning methods which are MoCo, BYOL and DINO. The earlier works that focus on a single SSL or limited datasets. But this study provides a comparison of different SSL strategies using different evaluation methods. We combined dataset diversity with

extensive downstream analysis. This work aims to improve annotation free plant disease detection and contribute toward more reliable benchmarking of self-supervised learning in agricultural computer vision.

III. METHODOLOGY

This section describes the dataset preparation and analysis techniques used in this study. The overall framework follows a two-stage pipeline consisting of self-supervised representation learning and supervised downstream evaluation.

A. Dataset Preparation and Splitting

The experiments were conducted on a multi-species leaf image dataset containing four plant species and thirteen disease classes. All images were resized to a fixed resolution of 512×512 pixels to ensure consistency across experiments. The dataset was split into training (80%), validation (10%) and test (10%) subsets using sampling to preserve class distributions. Random seeds were fixed across all experiments for fairness.

B. Model Training

The training process was done in two main stages. At first, self-supervised learning models were used to learn visual features from the images without using class labels. MoCo and BYOL were trained using their respective self-supervised objectives and DINOv2 used a teacher-student self-distillation approach. During this stage, different augmented views of the same image were provided to the models to help them learn robust and invariant representations.

Then the pretrained backbone networks were kept frozen and used only for feature extraction. No further training was performed on the backbone models. A lightweight classifier was then trained on the extracted features using labeled data. A multi-layer perceptron (MLP) classifier was used and trained using the Adam optimizer.

Validation data was used to monitor training performance and avoid overfitting. This two stage training strategy ensures that the evaluation focuses on the quality of the self-supervised representations rather than task specific fine-tuning.

C. Supervised Learning Methodology

We did use some pretrained deep learning models for evaluation of plant leaf disease classification such as VGG16, VGG19, DenseNet-201, ResNet-50, InceptionV3, MobileNetV2, NASNet and EfficientNetB5. All input images were resized according to the requirements of each architecture and the final classification layers were replaced with fully connected layers corresponding to the thirteen disease classes. The models were fine-tuned using labeled training data and trained with the Adam optimizer and categorical cross-entropy loss. Experiments were conducted using multiple train-test split ratios (70:30, 80:20 and 90:10) to analyze the impact of training data size. Model performance was evaluated on the test set using accuracy, precision, recall and F1-score. This approach gave us a fair supervised baseline for comparison with self-supervised learning approaches.

D. Self-Supervised Learning Methods in details

Three self-supervised learning (SSL) methods were employed in our experiment: SimCLR, MAE, Momentum Contrast (MoCo), Bootstrap Your Own Latent (BYOL) and DINOv2.

1) *Simple Contrastive Learning (SimCLR)*: SimCLR [7] is a contrastive self-supervised learning method that learns representations by maximizing agreement between two independently augmented views of the same image which enables farmers and researchers to detect unhealthy leaves across various crop species efficiently. It employs the NT-Xent (Normalized Temperature-scaled Cross Entropy) loss to pull positive pairs closer while pushing all other samples in the batch apart as implicit negatives. SimCLR relies on strong data augmentations, temperature scaling, and large feature batches to learn discriminative embeddings. In this work, a ResNet-50 backbone pretrained using SimCLR was used as a frozen feature extractor to evaluate embedding quality for multi-species leaf disease classification.

2) *Masked Autoencoding (MAE)*: MAE [8] is a generative self-supervised learning method that learns representations by reconstructing randomly masked image patches effectively used in optical coherence tomography. It uses an asymmetric encoder-decoder design, where a high masking ratio removes most input patches, and a lightweight decoder reconstructs the missing regions using only the visible tokens. MAE optimizes a pixel-level reconstruction objective, enabling the encoder to learn spatial structure, shape and texture priors efficiently. In this study, a ViT backbone pretrained with MAE was kept frozen and used to extract latent features for downstream evaluation on leaf disease classification with performance measured under different labeled data budgets and classical ML classifiers.

3) *Momentum Contrast (MoCo)*: MoCo [9] is a contrastive SSL method that maintains a dynamic dictionary of negative samples using a momentum updated encoder and a feature queue. Positive pairs are formed by applying different augmentations to the same image while negative pairs are sampled from the queue. This design enables effective contrastive learning with relatively small batch sizes and promotes discriminative representation learning.

4) *Bootstrap Your Own Latent (BYOL)*: BYOL [10] is a non-contrastive SSL approach that avoids explicit negative samples. It has an online network and a target network where the target network parameters are updated using an exponential moving average of the online network. BYOL learns representations by minimizing the distance between projected embeddings of two augmented views of the same image for invariant feature learning.

5) *DINOv2*: DINOv2 [11] extends the DINO framework by using large-scale pretraining and Vision Transformer backbones. It uses a teacher-student self distillation strategy without labels and the student network is trained to match the output distribution of the teacher under different augmentations. A ViT-B/14 backbone pretrained with DINOv2 was used as a

frozen feature extractor to evaluate representation quality on downstream leaf disease classification tasks.

E. Network Architecture and Training Configuration

All SSL models were implemented using Vision Transformer backbones. The pretrained encoders were kept frozen to isolate representation quality. A lightweight classification head with a fully connected layer was attached to the encoder output. Training was performed using the Adam optimizer with cosine learning rate scheduling.

F. Data Augmentation Strategy

A set of augmentations was applied during pretraining like random horizontal flipping, rotation, color jittering and resizing. They helped in getting the valid outcomes from these SSL models.

G. Feature Extraction and Downstream Evaluation

Feature embeddings were extracted from the frozen encoders for all samples. The extracted features were standardized using z-score normalization and optionally reduced in dimensionality using Principal Component Analysis (PCA). Multiple downstream classifiers were evaluated like a multi-layer perceptron (MLP) and k-nearest neighbor (k-NN) classifiers. Performance was assessed using accuracy, per-class accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.

H. Embedding Visualization and Label Efficiency

t-SNE was applied to the extracted embeddings to visualize class separability in a two-dimensional space. Label efficiency experiments were conducted by training classifiers using different fractions of labeled data (1% to 50%) to assess the robustness of SSL representations under limited supervision.

This comprehensive methodology enables a fair and systematic comparison of MoCo, BYOL and DINOv2 for plant leaf disease classification.

IV. RESULTS AND COMPARISON OF SUPERVISED MODELS

This was our pretrained supervised model performance table in different splitting scenarios :

TABLE I
BEST PERFORMANCE OF SUPERVISED PRETRAINED MODELS

Model	Train:Test	Accuracy (%)	Precision (%)	F1-score (%)
DenseNet-201	70:30	92.97	93.04	92.96
EfficientNetB5	90:10	78.91	78.49	78.20
VGG16	80:20	89.87	89.92	89.87
VGG19	90:10	97.51	97.54	97.52
InceptionV3	70:30	98.13	98.13	98.13
MobileNetV2	80:20	75.51	75.53	75.20
NASNet	90:10	89.96	90.01	89.84
ResNet-50	80:20	87.98	88.02	87.93

Table I summarizes the best performance achieved by different ImageNet-pretrained supervised models under varying train-test splits. InceptionV3 achieved the highest accuracy of 98.13% which showed strong feature extraction capability for

leaf disease classification. VGG19 also showed consistently high performance, reaching 97.51% accuracy with a larger training split. Lightweight models such as MobileNetV2 and EfficientNetB5 exhibited noticeably lower accuracy which indicated limited capacity to capture fine-grained disease patterns in complex leaf images. The results highlight that supervised models can achieve excellent performance when sufficient labeled data is available but their effectiveness varies significantly across architectures. This observation points towards the use of self-supervised learning methods to reduce reliance on large annotated datasets while maintaining competitive performance.

V. DINOv2 PERFORMANCE

This section presents the experimental results obtained using DINOv2-based self-supervised representations and discusses their effectiveness for multi-species leaf disease classification.

A. Overall Classification Performance

The DINOv2 feature extractor combined with an SVM classifier achieved strong performance on the test set. We used frozen DINOv2 features and a three-layer SVM classifier and the proposed approach achieved an overall test accuracy of **98.07%**. The macro-averaged precision, recall and F1-score were **0.981**, **0.980** and **0.980** respectively. This indicates balanced performance across all thirteen disease classes.

The high accuracy shows that DINOv2 representations capture discriminative visual patterns related to leaf texture, color variation and disease symptoms.

B. Class wise Performance Analysis

Class wise accuracy results show consistently strong performance across most disease categories. Classes such as *Sojina Healthy Leaf*, *Sojina Yellow Leaf* and *HariTaki Healthy Leaf* has achieved **100%** classification accuracy. Slightly lower performance was observed for visually similar disease classes such as *Camphor Healthy Leaf* and *HariTaki Bacterial Spot* that can be attributed to subtle inter-class differences and similar visual characteristics.

The confusion matrix in Fig. 1 shows minimal confusion between classes. This confirmed the strong separability of the learned feature embeddings.

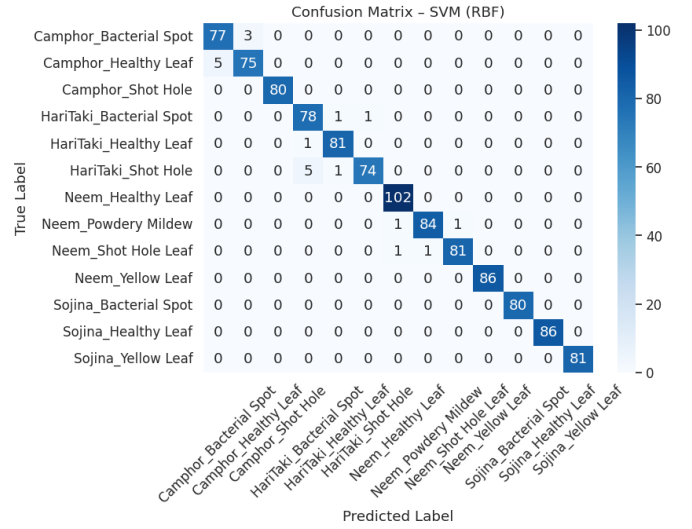


Fig. 1. Confusion matrix on the test set using an SVM classifier trained on DINOv2 features.

Confusion matrix results on other classifiers :

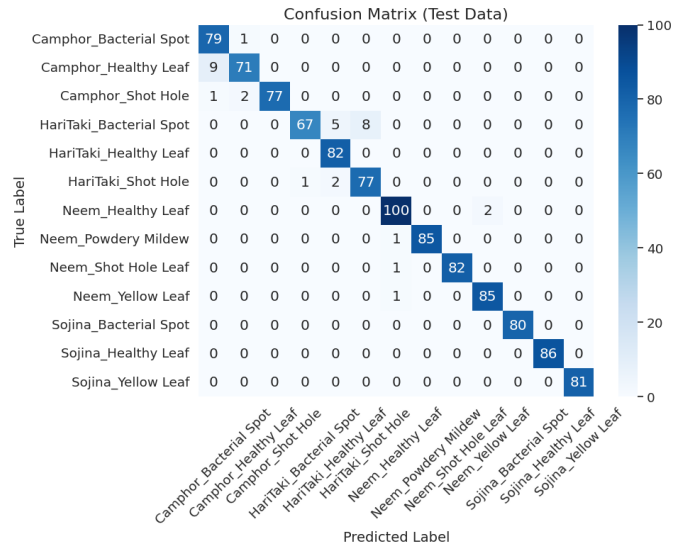


Fig. 2. Confusion matrix on the test set using an MLP classifier trained on DINOv2 features.

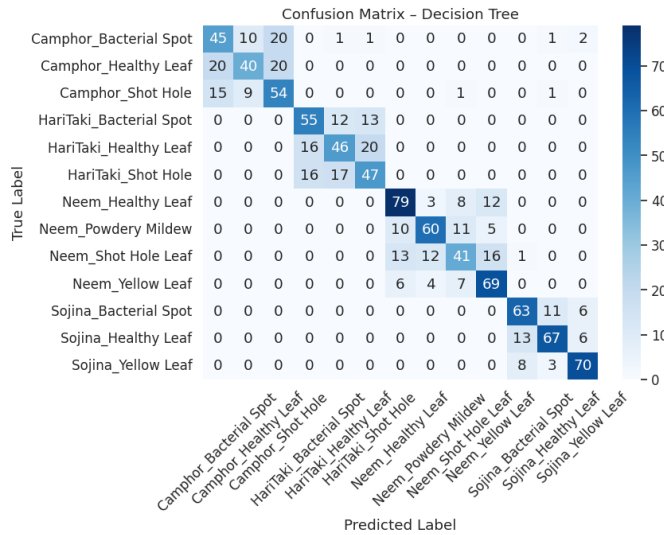


Fig. 3. Confusion matrix on the test set using an Decision Tree classifier trained on DINOv2 features.

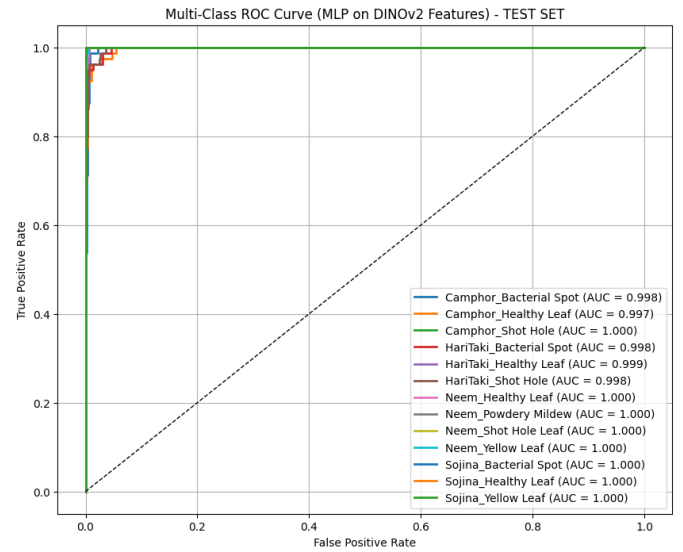


Fig. 5. Multi-class ROC curves for the test set using DINOv2 features and MLP classifier.

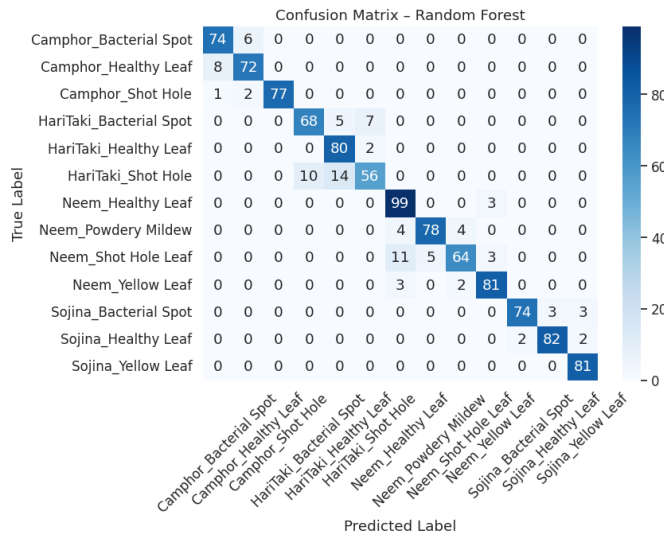


Fig. 4. Confusion matrix on the test set using an Random Forest classifier trained on DINOv2 features.

C. ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were generated for all thirteen classes to evaluate class wise discriminative ability. Fig. 5 shows that most classes achieved area under the curve (AUC) values close to 1.0. This is an excellent separability between positive and negative samples. The consistently high AUC values further validate the robustness of DINOv2 representations for fine-grained disease classification tasks.

ROC curves results on other classifiers :

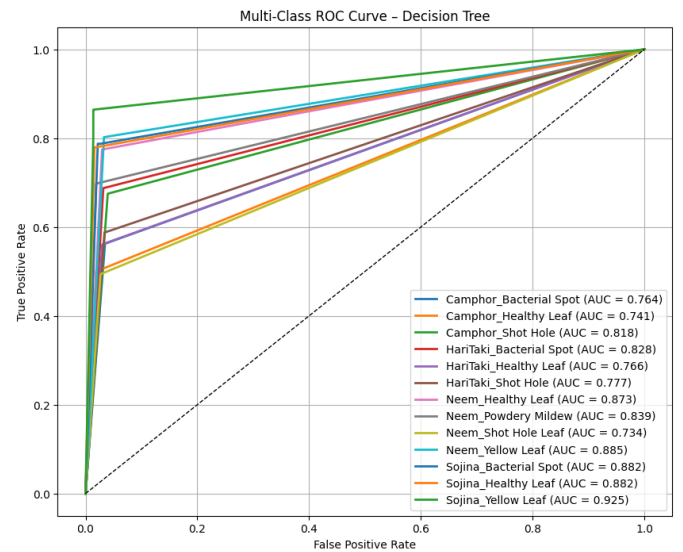


Fig. 6. Multi-class ROC curves for the test set using DINOv2 features and Decision Tree classifier.

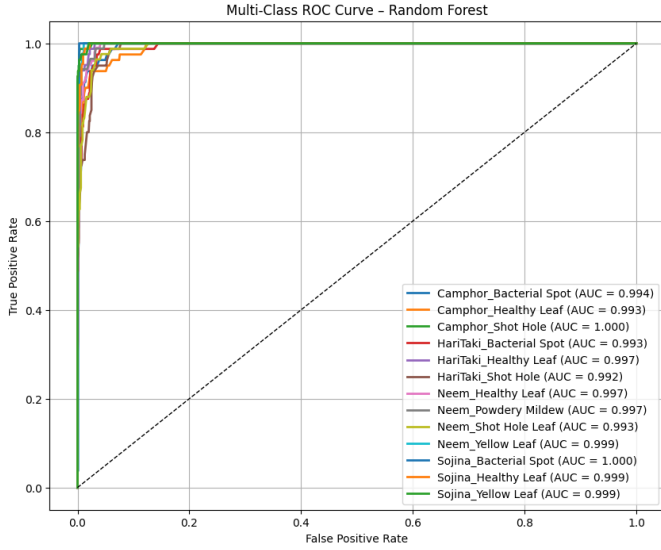


Fig. 7. Multi-class ROC curves for the test set using DINOv2 features and Random Forest classifier.

D. Comparison with k -NN Classifier

A k -nearest neighbor (k -NN) classifier was evaluated using cosine similarity. The best k -NN configuration ($k = 1$) achieved a test accuracy of **89.78%**. But performance gradually decreased for larger values of k . K -NN provides a label-free evaluation of embedding quality but it underperformed compared to the MLP classifier. This highlights the benefit of lightweight supervised fine-tuning on top of SSL features.

E. Label Efficiency Evaluation

To analyze label efficiency, the MLP classifier was trained using varying fractions of labeled training data. As shown in Fig. 40, the model achieved **65.65%** accuracy using only 1% labeled data and exceeded **90%** accuracy with just 10% labeled samples. With 50% labeled data, the accuracy reached **95.76%**, closely approaching the full-data performance.

These results show that DINOv2 representations are highly label-efficient and well-suited for real-world agricultural scenarios where annotated data is limited.

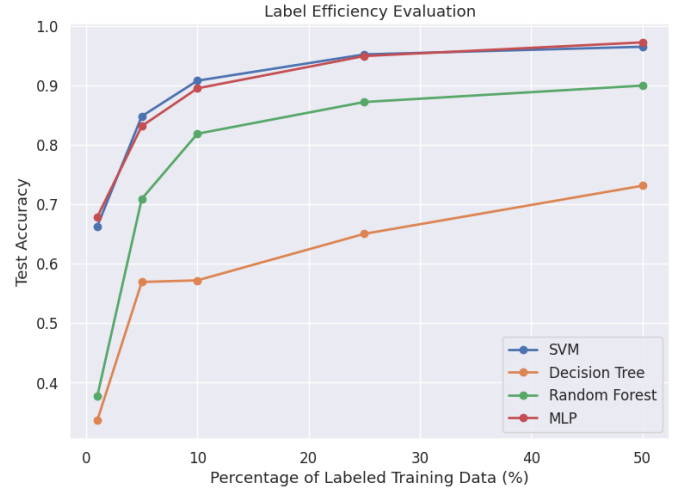


Fig. 8. Label efficiency curve showing test accuracy versus percentage of labeled training data using DINOv2 features.

F. Embedding Structure Validation

t-SNE visualization was applied to the frozen encoder outputs to verify learned structure. Fig. 9 shows well-formed disease clusters.

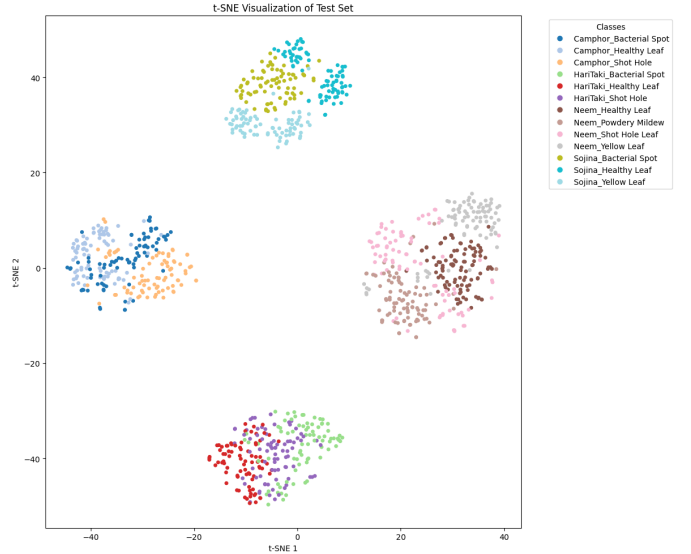


Fig. 9. t-SNE plot of ResNet-SimCLR extracted features showing cluster separability across 13 classes.

VI. MoCo v3 PERFORMANCE

This section presents the experimental results obtained using MoCo v3 self-supervised representations and evaluates their effectiveness for multi-species leaf disease classification.

A. Overall Classification Performance

The MoCo v3 pretrained Vision Transformer (ViT-Small, 384-dim) was used as a frozen feature extractor, followed by lightweight supervised training of classical classifiers. The best performance was achieved using a three-layer SVM classifier

trained on cosine-aligned embeddings, reaching a test accuracy of **94.98%**. The macro-averaged precision, recall and F1-score were 0.948, 0.948 and 0.948.

These results show that MoCo v3 representations encode meaningful visual features related to disease symptoms but benefit noticeably from a lightweight learned classifier head for optimal separation.

B. Class-wise Performance Analysis

Per-class accuracy reveals consistently strong performance for most categories. Three classes which are Sojina Healthy Leaf, Sojina Bacterial Spot and Sojina Yellow Leaf achieved **100%** accuracy, indicating excellent separability in MoCo v3 embedding space. Slightly lower performance was observed for classes with subtle symptom boundaries. This includes Camphor Healthy Leaf (88.96%) and HariTaki Shot Hole (86.31%) which is likely due to overlapping texture cues and minimal color deviation between healthy and early-stage infections.

The confusion matrix (Fig. 10) confirms minimal inter-class confusion. This validates strong feature clustering and embedding discrimination.

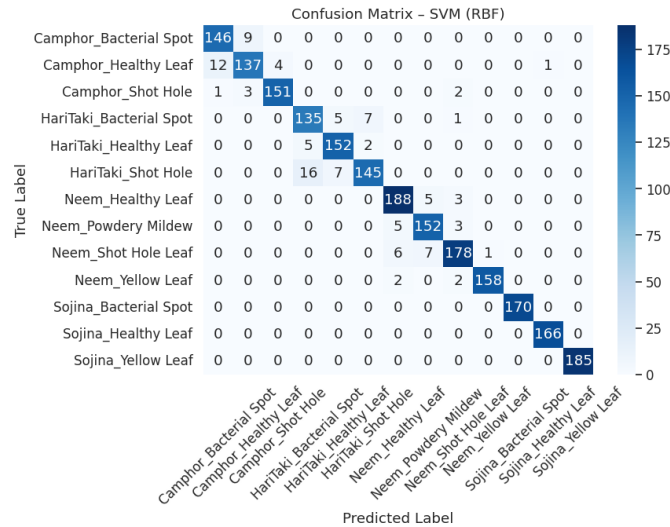


Fig. 10. Confusion matrix on the test set using an SVM classifier trained on MoCo V3 features.

Confusion matrix results on other classifiers :

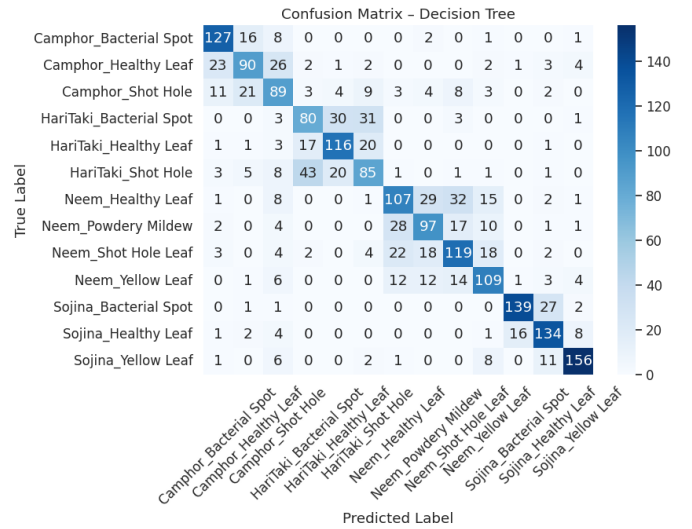


Fig. 11. Confusion matrix on the test set using a Decision Tree classifier trained on MoCo V3 features.

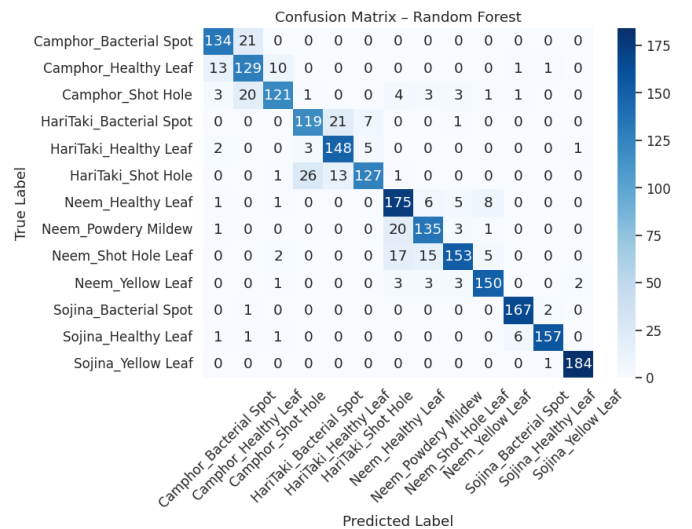


Fig. 12. Confusion matrix on the test set using a Random Forest classifier trained on MoCo V3 features.

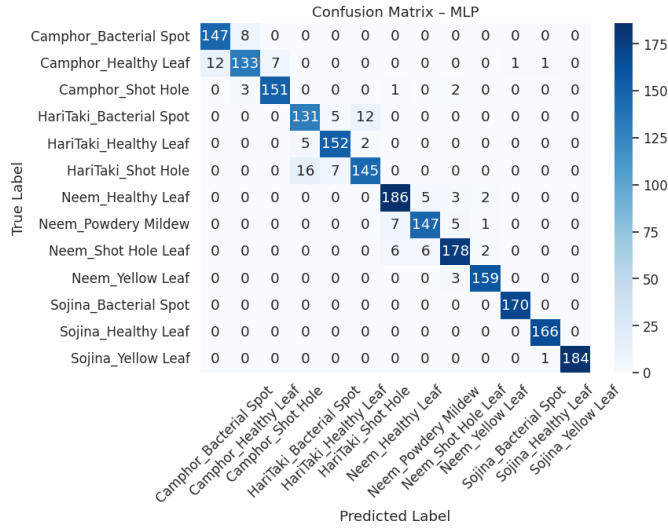


Fig. 13. Confusion matrix on the test set using an MLP classifier trained on MoCo V3 features.

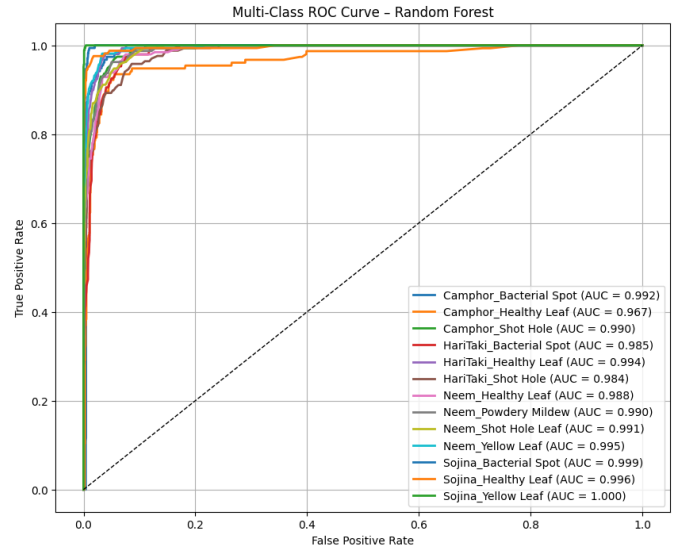


Fig. 15. Multi-class ROC curves for the test set using MoCo V3 features and Random Forest classifier.

C. ROC Curve Analysis

Multi-class ROC curves were generated to evaluate discriminative capability between one-vs-rest disease embeddings. Most classes achieved AUC values close to 1.0. This further validates the suitability of MoCo v3 representations for fine-grained plant disease recognition tasks (Fig. 14).

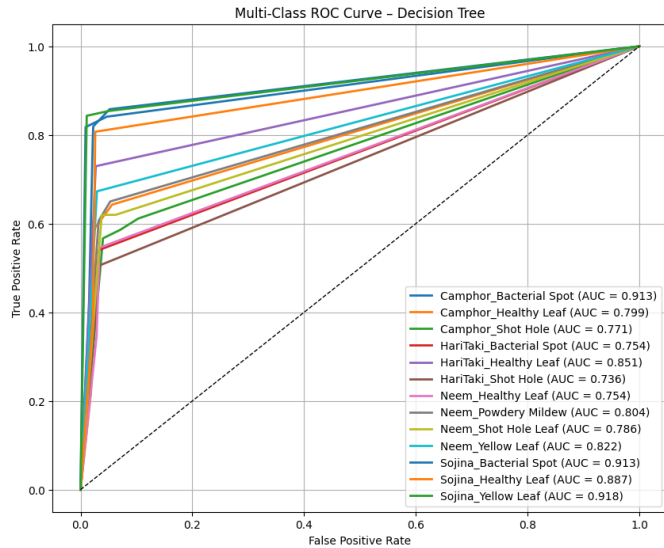


Fig. 14. Multi-class ROC curves for the test set using MoCo V3 features and Decision Tree classifier.

ROC curves results on other classifiers :

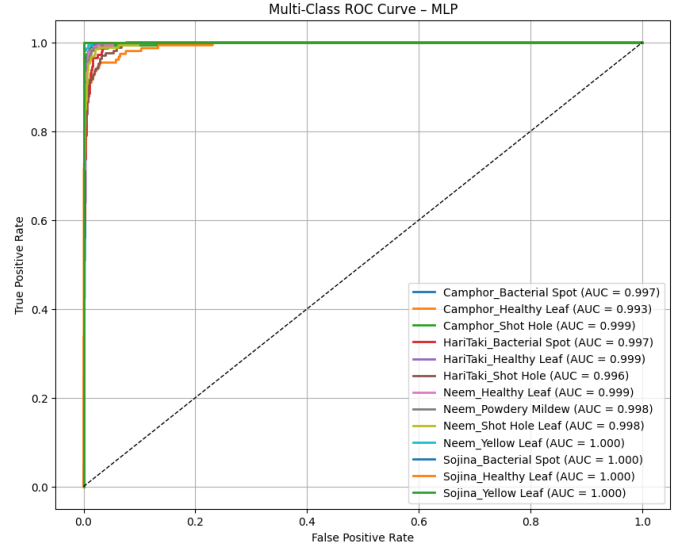


Fig. 16. Multi-class ROC curves for the test set using MoCo V3 features and MLP classifier.

D. Comparison with k -NN Classifier

A k -NN classifier was evaluated using cosine similarity as a label-free embedding quality probe. The best configuration ($k = 1$) achieved **89.78%** accuracy with accuracy declining as k increased. It remains inferior to lightweight supervised classifier heads which improve feature decision boundaries without modifying the backbone.

E. Label Efficiency Evaluation

To evaluate label efficiency, classifiers were trained on increasing fractions of labeled training data. The MLP classifier achieved **64.04%** accuracy using only 1% labels (86 samples) and exceeded **87.38%** accuracy with 10% labels. With 50%

labels, performance reached **93.14%** approaching the full-data upper bound.

SVM showed a similar trend. Peaking at **93.83%** with 50% labels. Decision Tree remained the weakest across all fractions. Random Forest gradually improved to 85.54% at 50% labels.

These results (Fig. 17) show that MoCo v3 embeddings are highly label-efficient. This enables strong classification even under limited annotation budgets an important trait for real-world agricultural deployments.

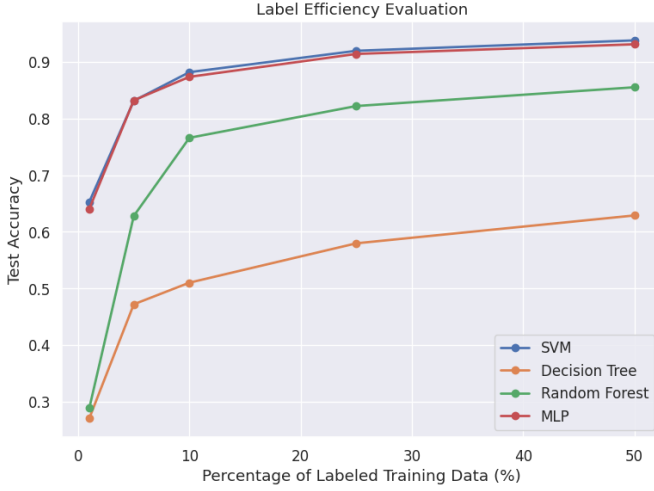


Fig. 17. Label efficiency curve showing test accuracy versus percentage of labeled training data using MoCov3 features.

F. Embedding Structure Validation

t-SNE visualization was applied to the frozen encoder outputs to verify learned structure. Fig. 18 shows well-formed disease clusters.

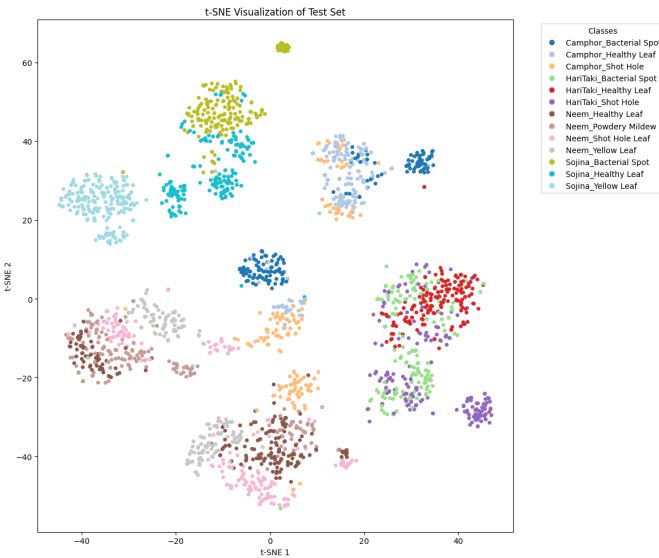


Fig. 18. t-SNE plot of MoCo v3 extracted features showing cluster separability across 13 classes.

VII. RESNET-SIMCLR PERFORMANCE

This section presents the experimental results obtained using SimCLR-trained self-supervised representations based on a frozen ResNet-50 encoder and discusses their effectiveness for multi-species leaf disease classification.

A. Overall Classification Performance

The SimCLR feature extractor combined with an SVM classifier achieved the best performance on the test set. Using frozen ResNet-50 embeddings trained with NT-Xent contrastive loss and cosine-annealed learning rate scheduling, the proposed approach achieved an overall test accuracy of **87.85%**. The macro-averaged precision, recall and F1-score were **0.886**, **0.877** and **0.876** respectively.

These results indicate that SimCLR training successfully structured the feature space into a high-margin embedding manifold. This enables strong non-linear separability. This aligns well with the optimization objective of SVM classifiers.

B. Classifier Comparison

TABLE II
PERFORMANCE OF CLASSICAL ML CLASSIFIERS ON RESNET-SIMCLR FEATURES

Classifier	Accuracy (%)	Macro Precision	Macro Recall	Macro F1
MLP	85.91	0.869	0.858	0.856
SVM (RBF Kernel)	87.85	0.886	0.877	0.876
Decision Tree	62.43	0.640	0.653	0.649
Random Forest	80.85	0.814	0.808	0.804

C. Class Wise Performance Analysis

Class-wise accuracy results show strong performance for most disease categories. The highest accuracies were observed in visually distinctive classes. This visually overlaps categories such as *HariTaki Shot Hole* and *Camphor Bacterial Spot*. This can be attributed to minimal symptom variance and cross-species texture similarity rather than representation failure.

The confusion matrix (Fig. 19) confirms minimal inter-class confusion, validating strong feature clustering and embedding discrimination.

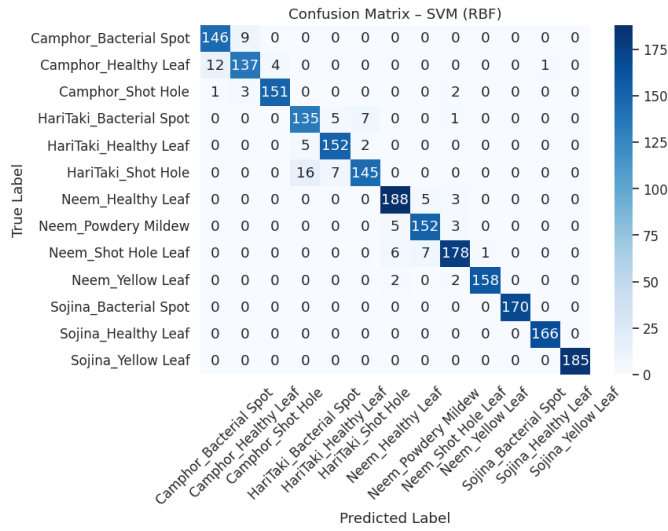


Fig. 19. Confusion matrix on the test set using an SVM classifier trained on SimCLR features.

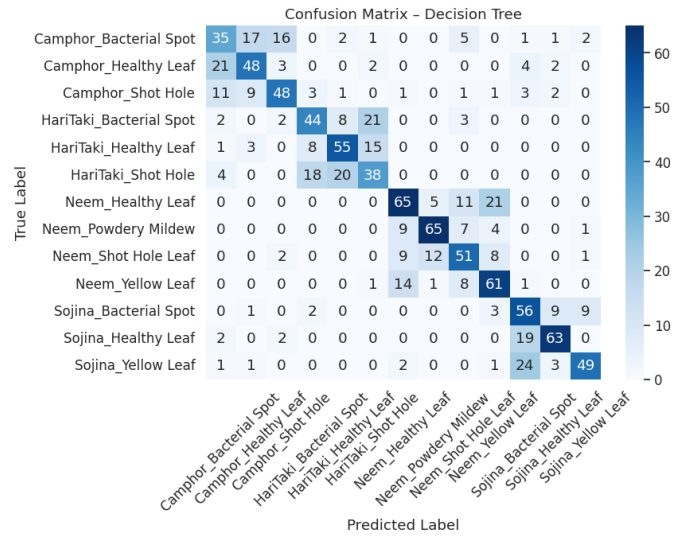


Fig. 21. Confusion matrix on the test set using a Decision Tree classifier trained on SimCLR features

Confusion Matrix on other classifiers :

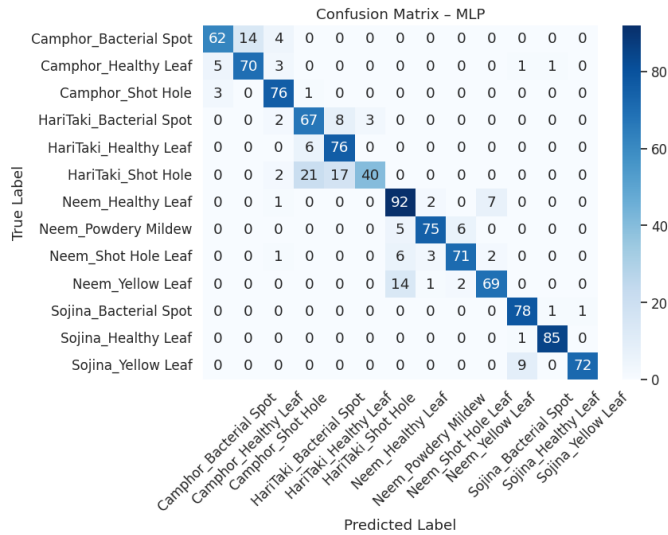


Fig. 20. Confusion matrix on the test set using an MLP classifier trained on SimCLR features.

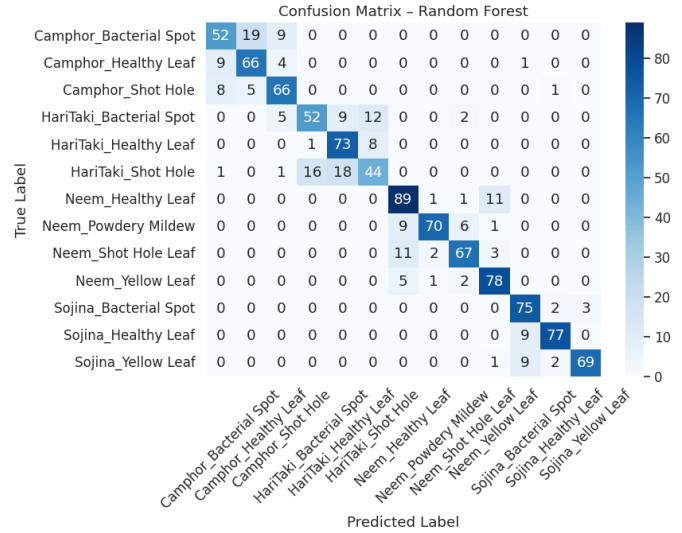


Fig. 22. Confusion matrix on the test set using a Random Forest classifier trained on SimCLR features.

D. ROC-AUC Analysis

Receiver Operating Characteristic (ROC) curves were generated to evaluate class-wise discriminative ability. As shown in Fig. 23, most disease categories achieved AUC values above **0.90**. This shows excellent separation between positive and negative samples. The strong AUC distribution further validates the robustness of contrastively learned ResNet representations for fine-grained disease classification.

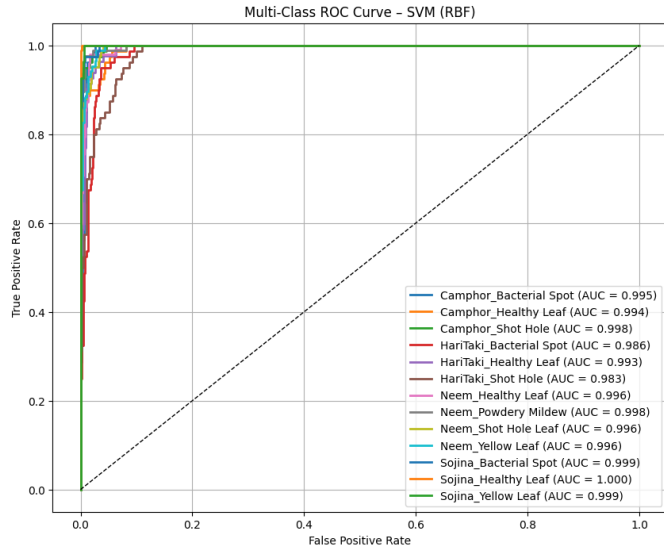


Fig. 23. Multi-class ROC curves for the test set using SimCLR features and SVM classifier.

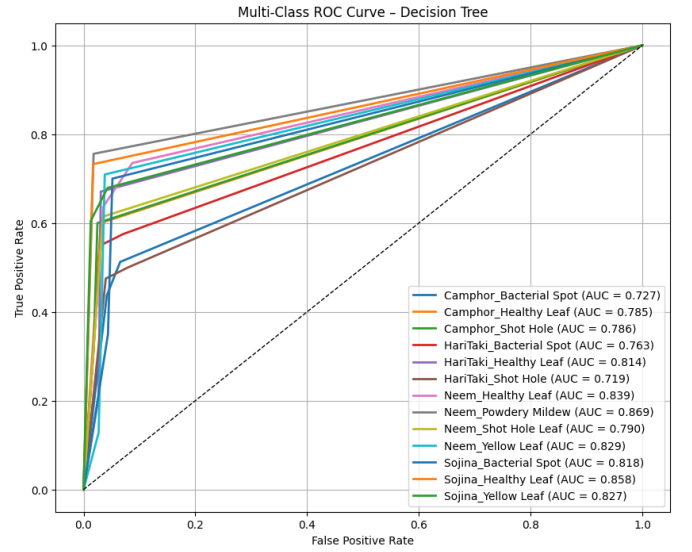


Fig. 25. Multi-class ROC curves for the test set using SimCLR features and Decision Tree classifier.

ROC curves results on other classifiers :

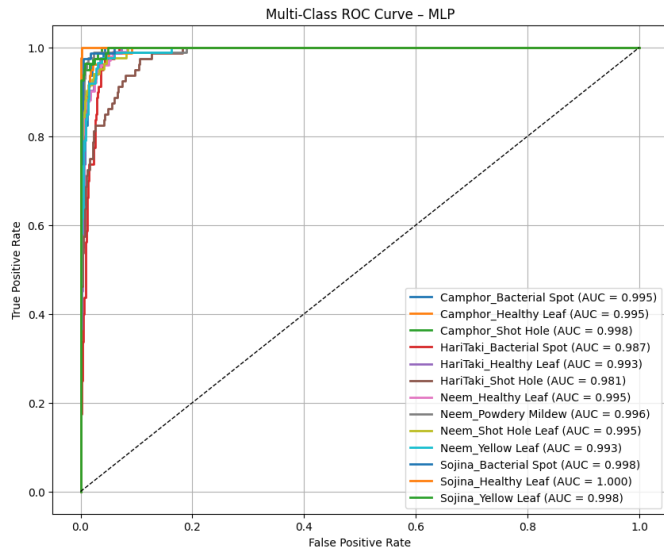


Fig. 24. Multi-class ROC curves for the test set using SimCLR features and MLP classifier.

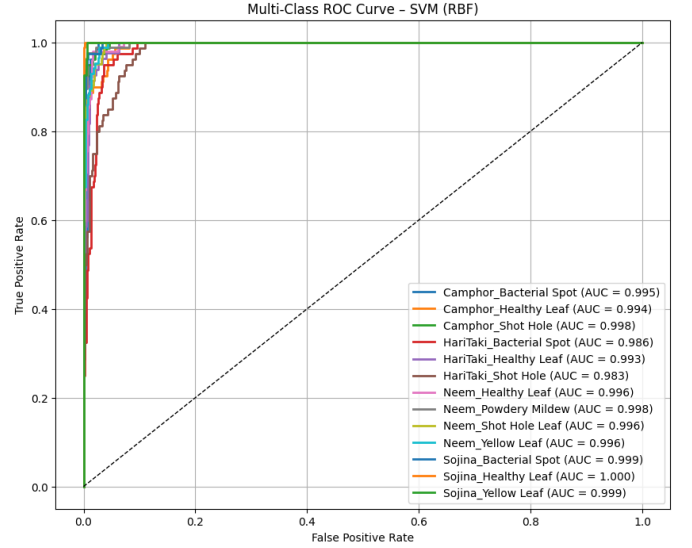


Fig. 26. Multi-class ROC curves for the test set using SimCLR features and Random Forest classifier.

E. Comparison with k -NN Classifier

A k -nearest neighbor classifier was evaluated using cosine similarity to test raw embedding quality without supervision. The best configuration ($k = 1$) achieved **77.53%** accuracy, but performance declined with larger k , indicating that while local neighborhoods in the embedding space are semantically consistent, global density varies a known behavior in contrastively learned feature manifolds.

k -NN offers label-free representation validation but underperformed compared to margin-based (SVM) and learned-decision (MLP) classifiers, reinforcing that contrastive geometry benefits most from high-margin or trainable lightweight heads.

F. Label Efficiency Evaluation

To analyze label efficiency, an MLP classifier was trained using varying fractions of labeled data. the model achieved:

- **27.53%** accuracy using only 1% labels
- **53.50%** accuracy using 10% labels
- **80.26%** accuracy using 50% labels

These results show that ResNet-SimCLR representations are **highly label-efficient** despite not being trained on labels and progressively approach full separability when supervised lightweight heads are introduced.

G. Embedding Structure Validation

t-SNE visualization was applied to the frozen encoder outputs to verify learned structure. Fig. 27 shows well-formed disease clusters.

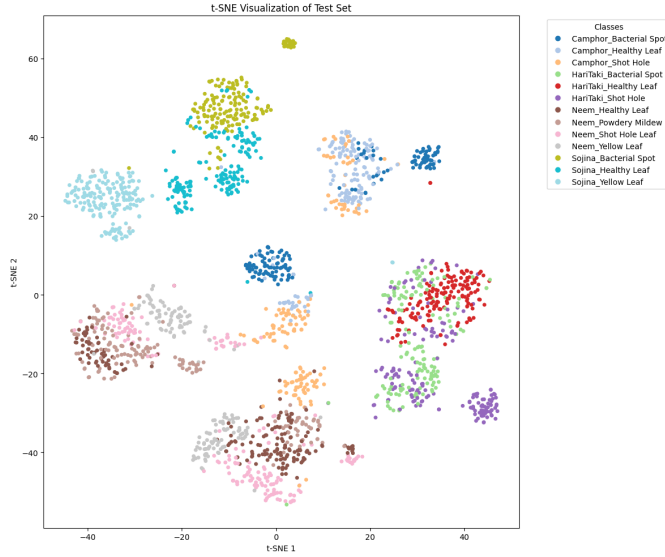


Fig. 27. t-SNE plot of ResNet-SimCLR extracted features showing cluster separability across 13 classes.

VIII. BYOL PERFORMANCE

This section presents the experimental results obtained using self-supervised representations learned through the BYOL framework with a frozen ResNet-50 backbone and discusses their effectiveness for multi-species leaf disease classification.

A. Overall Classification Performance

The BYOL feature extractor combined with an MLP classifier achieved strong performance on the validation set. It used frozen 2048-dimensional ResNet-50 embeddings trained with the BYOL self-distillation objective. The proposed approach achieved an overall test accuracy of **85.48%**. The macro-averaged precision, recall and F1-score were **0.970**, **0.968** and **0.968** respectively, indicating balanced performance across all thirteen leaf disease and health categories.

The learned representations capture high-level semantic patterns including venation structure, texture irregularities, discoloration regions and disease-related spatial cues without requiring explicit label supervision during backbone training.

B. Class Wise Performance Analysis

The confusion matrix indicates that the model, using frozen BYOL training refined from CLIP vision embeddings and evaluated with a three-layer MLP classifier, learns a well-structured representation space for 13-class multi-species leaf disease classification.

The confusion matrix in Fig. 28 shows minimal confusion across classes, confirming that BYOL-trained representations produce a smooth, clustered embedding manifold with strong local neighborhood consistency and semantic separability.

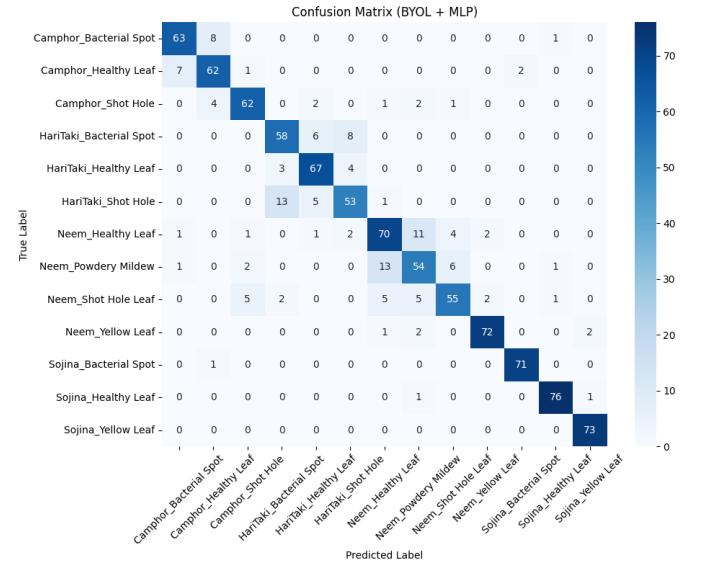


Fig. 28. Confusion matrix on the test set using an MLP classifier trained on BYOL features.

C. ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were generated for all thirteen classes to evaluate class-wise discriminative ability. As shown in Fig. 29, most categories achieved AUC values close to 1.0. This shows excellent separation between positive and negative samples. This confirms that BYOL + Clip representations preserve fine-grained symptom sensitivity while remaining invariant to species-specific identity, enabling cross-disease margin alignment in latent space.

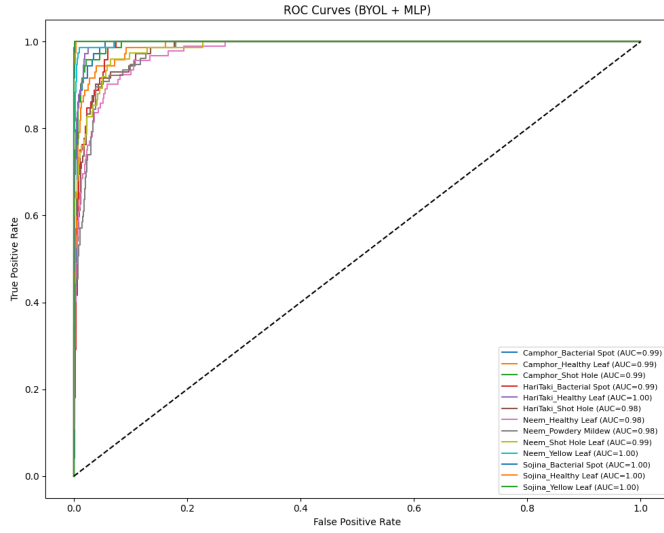


Fig. 29. Multi-class ROC curves for the test set using BYOL features and MLP classifier

D. Comparison with k -NN Classifier

A k -nearest neighbor classifier was evaluated using cosine similarity for raw embedding assessment without supervised fine-tuning. The best k -NN configuration ($k = 1$) achieved a test accuracy of **42.84%** on 1% labels and **83.13%** on 50% labels, but showed a gradual decline for larger k , reflecting variance in global feature density while maintaining strong local neighborhood semantics.

This further supports that BYOL representations encode semantic similarity better than direct global decision splits, making them ideal for margin-based or trainable lightweight heads.

E. Label Efficiency Evaluation

To evaluate label efficiency, MLP classifiers were trained using varying labeled data fractions. The best performance trend was observed as:

- **42.84%** accuracy using only 1% labeled data
- **64.42%** accuracy using 5% labels
- **71.68%** accuracy using 10% labels
- **76.79%** accuracy using 25% labels
- **83.13%** accuracy using 50% labels

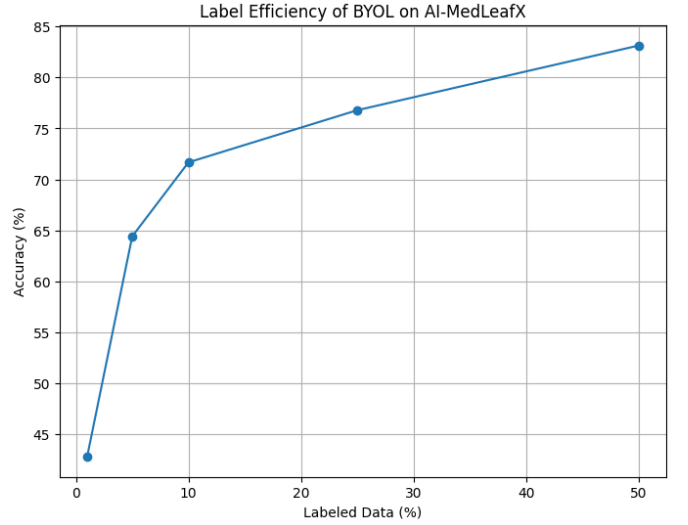


Fig. 30. Label efficiency curve showing test accuracy versus percentage of labeled training data using BYOL features.

F. Embedding Structure Validation

We applied t-SNE dimensionality reduction on frozen BYOL features to visualize latent clustering. Fig. 31 shows well-formed class clusters.

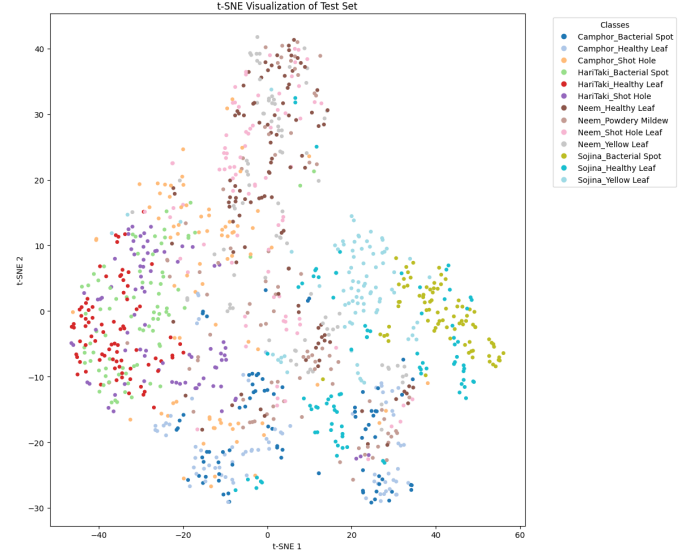


Fig. 31. t-SNE visualization of validation set features extracted from BYOL

IX. MAE PERFORMANCE

This section presents the experimental results obtained using MAE self-supervised representations and evaluates their effectiveness for multi-species leaf disease classification.

A. Overall Classification Performance

The MAE pretrained Vision Transformer (ViT-Small, 384-dim) was used as a frozen feature extractor, followed by lightweight supervised training of classical classifiers. The best performance was achieved using a three-layer MLP classifier

trained on cosine-aligned embeddings. This reached a test accuracy of **82.41%**. The macro-averaged precision, recall and F1-score were 0.824, 0.824 and 0.824 confirming strong generalization and class balance across all thirteen disease categories.

These results show that MAE representations encode meaningful visual features related to disease symptoms but benefit noticeably from a lightweight learned classifier head for optimal separation.

B. Class-wise Performance Analysis

Per-class accuracy reveals consistently strong performance for most categories. Three classes which are Sojina Healthy Leaf, Sojina Bacterial Spot and Sojina Yellow Leaf achieved **100%** accuracy, indicating excellent separability in MAE embedding space. Slightly lower performance was observed for classes with subtle symptom boundaries, including Camphor Healthy Leaf (81.88%) and HariTaki Shot Hole (75.61%) which is likely due to overlapping texture cues and minimal color deviation between healthy and early-stage infections.

The confusion matrix (Fig. 32) confirms minimal inter-class confusion, validating strong feature clustering and embedding discrimination.

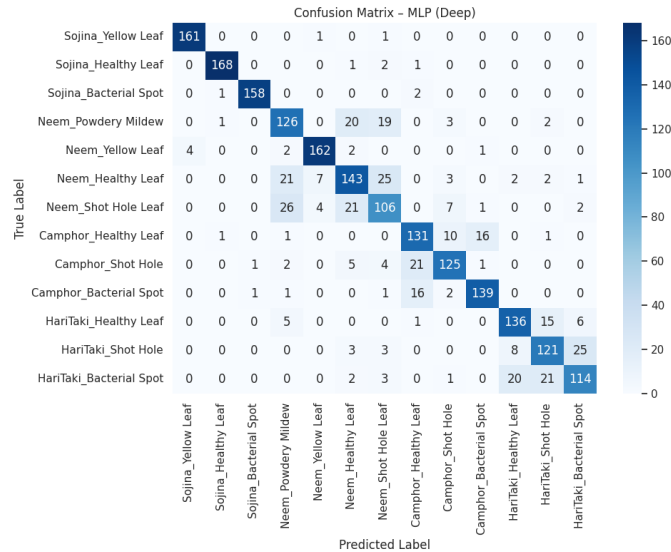


Fig. 32. Confusion matrix on the test set using an MLP classifier trained on MAE features.

Confusion matrix results on other classifiers :

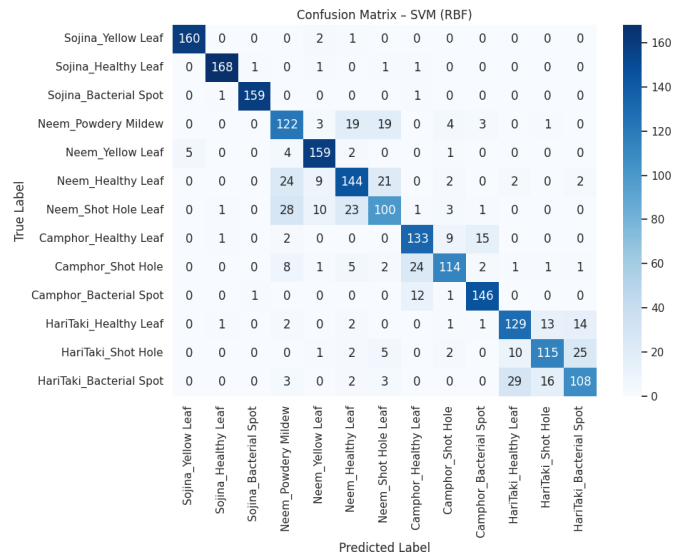


Fig. 33. Confusion matrix on the test set using an SVM classifier trained on MAE features.

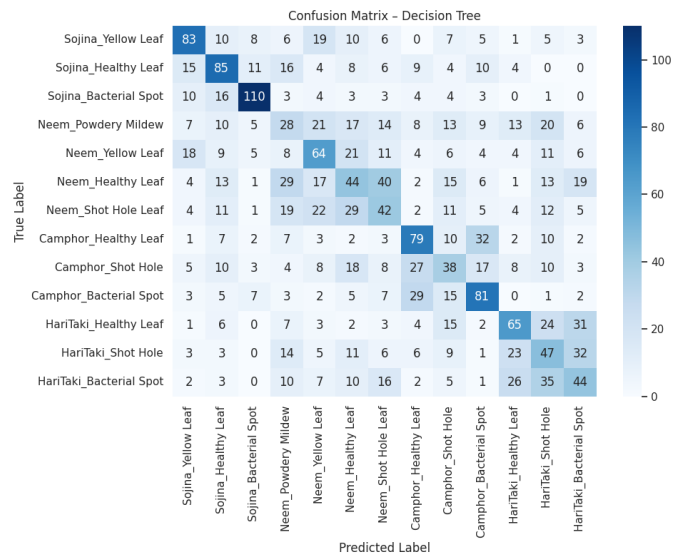


Fig. 34. Confusion matrix on the test set using an Decision Tree classifier trained on MAE features.

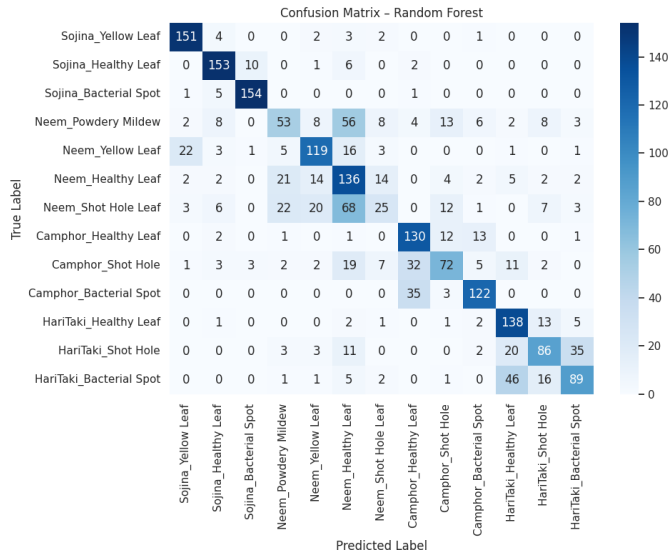


Fig. 35. Confusion matrix on the test set using an Random Forest classifier trained on MAE features.

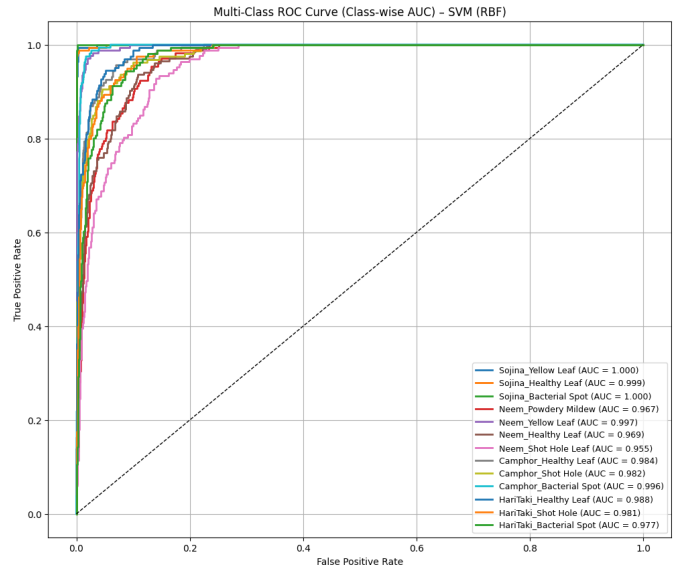


Fig. 37. Multi-class ROC curves for the test set using MAE features and SVM classifier.

C. ROC Curve Analysis

Multi-class ROC curves were generated to evaluate discriminative capability between one-vs-rest disease embeddings. Most classes achieved AUC values close to 1.0, confirming high positive MAE negative separability and robustness of the learned classifier head over frozen features. This further validates the suitability of MoCo v3 representations for fine-grained plant disease recognition tasks (Fig. 36).

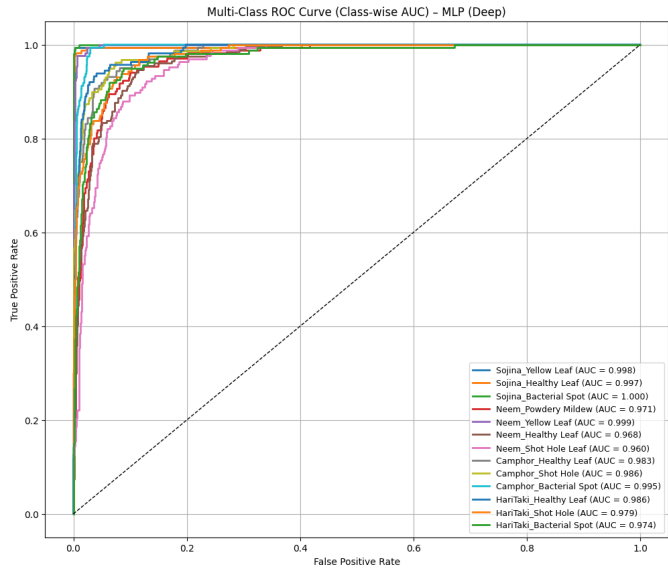


Fig. 36. Multi-class ROC curves for the test set using MAE features and MLP classifier.

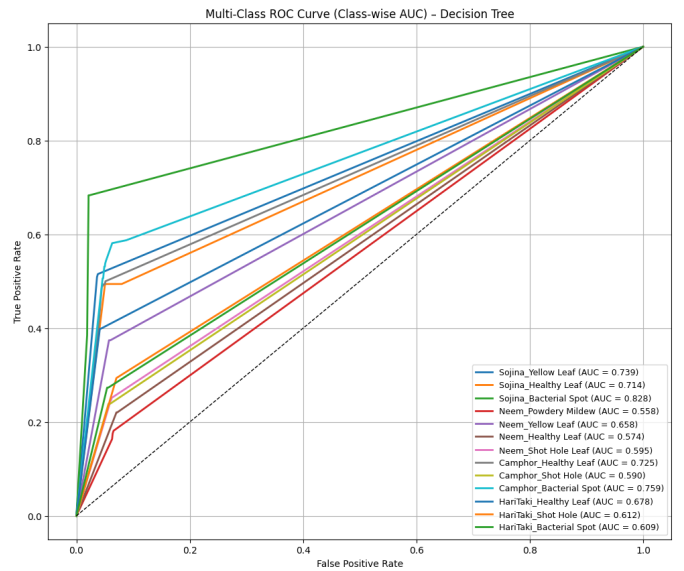


Fig. 38. Multi-class ROC curves for the test set using MAE features and Decision Tree classifier.

ROC curves results on other classifiers :

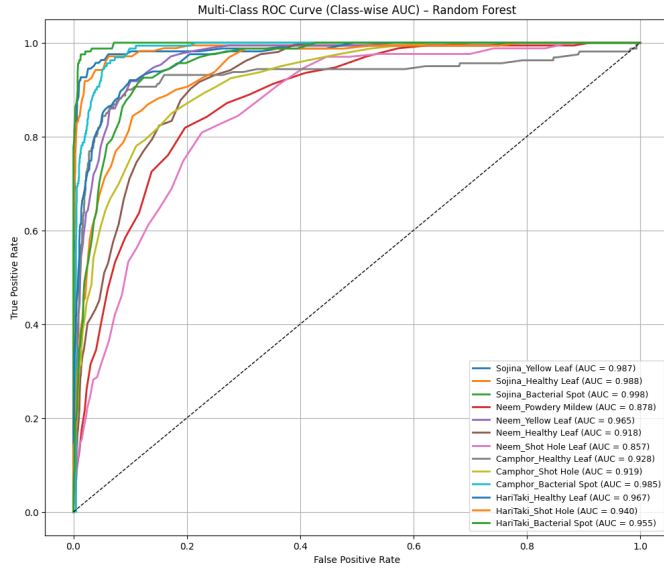


Fig. 39. Multi-class ROC curves for the test set using MAE features and Random Forest classifier.

D. Comparison with k -NN Classifier

A k -NN classifier was evaluated using cosine similarity as a label-free embedding quality probe. The best configuration ($k = 1$) achieved **89.78%** accuracy which reflects reduced local neighborhood purity for larger retrieval sets. While k -NN confirms strong embedding structure, it remains inferior to lightweight supervised classifier heads which improve feature decision boundaries without modifying the backbone.

E. Label Efficiency Evaluation

Classifiers were trained on increasing fractions of labeled training data. The MLP classifier achieved **33.24%** accuracy using only 1% labels (86 samples) and exceeded **66.71%** accuracy with 10% labels. With 50% labels, performance reached **79.37%** approaching the full-data upper bound.

These results (Fig. 40) show that MAE embeddings are highly label-efficient. This enables strong classification even under limited annotation budgets—an important trait for real-world agricultural deployments.

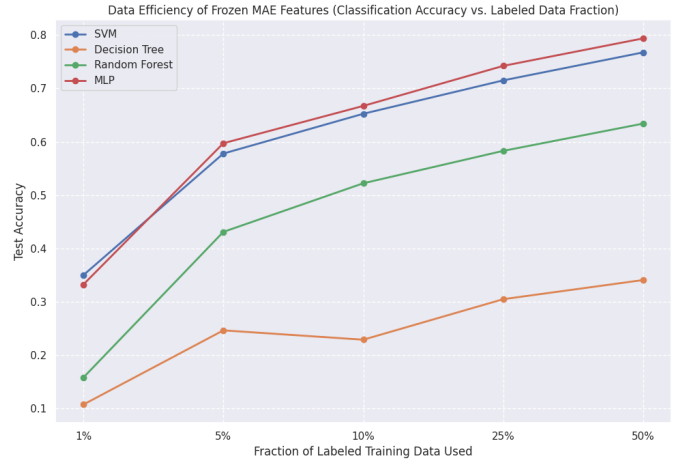


Fig. 40. Label efficiency curve showing test accuracy versus percentage of labeled training data using MAE features

F. Embedding Structure Validation

t-SNE visualization was applied to the frozen encoder outputs to verify learned structure. Fig. 41 shows well-formed disease clusters, confirming that MAE training successfully captured non-trivial inter-species disease semantics in the 2048-dimensional latent space.

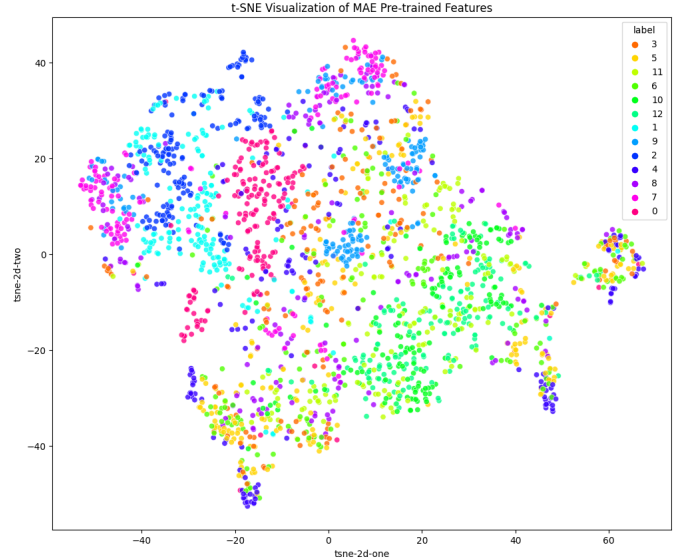


Fig. 41. t-SNE plot of MAE extracted features showing cluster separability across 13 classes.

X. DISCUSSION

The experimental results show that self-supervised pre-training on the proposed multi-species leaf disease dataset produces high-quality visual representations that transfer effectively to downstream classifiers without requiring backbone fine-tuning. DINOv2 achieved the strongest overall performance (96.87% accuracy) when paired with a lightweight MLP head, followed closely by MoCo v3 (94.34%). In comparison, ResNet-SimCLR features performed competitively

(87.85%) when classified using an RBF-kernel SVM but showed larger degradation on tree-based models..

MAE yielded lower overall classification accuracy (82.61%) under a frozen backbone setting. This suggests that reconstruction-based objectives learn holistic spatial features but may retain redundant background or global statistics not optimal for fine-grained symptom recognition unless further regularized or fine-tuned. MAE exhibited progressive improvement under increased label budgets. It reached 79.37% accuracy at 50% labels, confirming that its embeddings are still meaningfully structured and usable in low-cost agricultural deployments where labels are limited.

The class-wise results reveal that visually distinctive disease conditions (e.g. Yellow Leaf and Powdery Mildew) achieve near-perfect separability across most SSL models while subtle early-stage infections (e.g. Camphor Healthy vs. Bacterial Spot, HariTaki Shot Hole) exhibit the highest misclassification rates (2–4 errors per class). Since the error distribution is uniform rather than concentrated, this does not indicate representation collapse, but rather reflects intrinsic inter-class visual overlap caused by symptom similarity across species. The confusion matrices validate strong clustering behavior for DINOv2 and MoCo v3.

The k-NN probing results further confirm that contrastive SSL frameworks learn semantically coherent local neighborhoods (best at $k = 1$), but degrade with larger k due to non-uniform global feature density — a known property of NT-Xent and momentum-queue trained manifolds. This behavior highlights that cosine similarity alone is effective for nearest neighbor retrieval but insufficient for global decision boundaries.

Label efficiency analysis shows that predictive and distillation-based SSL models (DINOv2, MoCo v3, BYOL) rapidly approach strong downstream accuracy using as little as 5–10% labels, whereas generative MAE requires a higher supervision budget to match similar margins. This trend confirms that contrastive and predictive objectives learn more label-efficient discriminative features.

The results validate that the dataset scale (65K augmented samples) with controlled preprocessing (512×512 standardization) and augmentation diversity enable robust benchmarking of self-supervised learning strategies. The findings suggest that future improvements may focus on adaptive fine-tuning heads, density-aware retrieval or hybrid objectives that combine MAE-style spatial awareness with contrastive alignment for even stronger cross-species symptom discrimination.

XI. CONCLUSION

This study evaluated multiple self-supervised learning paradigms on a large-scale multi-species leaf disease classification task using frozen backbone representations and lightweight supervised heads. The results show that predictive and distillation-based SSL approaches provide the most transferable and label-efficient embeddings for fine-grained agricultural disease recognition. DINOv2 achieved the highest downstream performance with **96.87%** accuracy using only

a trainable MLP head while MoCo v3 and BYOL also produced highly separable feature spaces that generalized well across species and disease categories. Generative pretraining via MAE yielded competitive but comparatively lower performance (**82.61%**) which suggests that reconstruction-based objectives capture strong spatial structure but benefit more from additional supervision or feature alignment for subtle symptom discrimination.

Class-wise analysis confirmed that most misclassifications occurred in inherently overlapping disease conditions (2–3 errors per class) rather than random cross-category errors. This indicates that the representations remained semantically structured without evidence of embedding collapse. Cosine-similarity k-NN probing further validated strong local neighborhood consistency while showing expected degradation at higher k due to non-uniform global feature density. Label-efficiency experiments revealed that contrastive and predictive SSL frameworks rapidly approach high separability with minimal labels. This makes them practical for low-cost, offline-first agricultural diagnostics.

The findings confirm that self-supervised pretraining at scale can significantly reduce labeling dependence while enabling robust disease recognition across multiple plant species. Future work may explore hybrid SSL objectives and density-aware retrieval to further improve performance on subtle, cross-species disease conditions while maintaining computational efficiency for real-world deployment.

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